

Public Key Crypto notes - 09

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1 PoK protocols we learnt about

Recall 1.1 (Proof-of-knowledge protocols).

1. $\text{PoK}[x : g^x = h](m) \leftarrow \text{Schnorr signature}$
2. $\text{PoK}[x : g^{a^x} = y](m)$, g, a, y are public

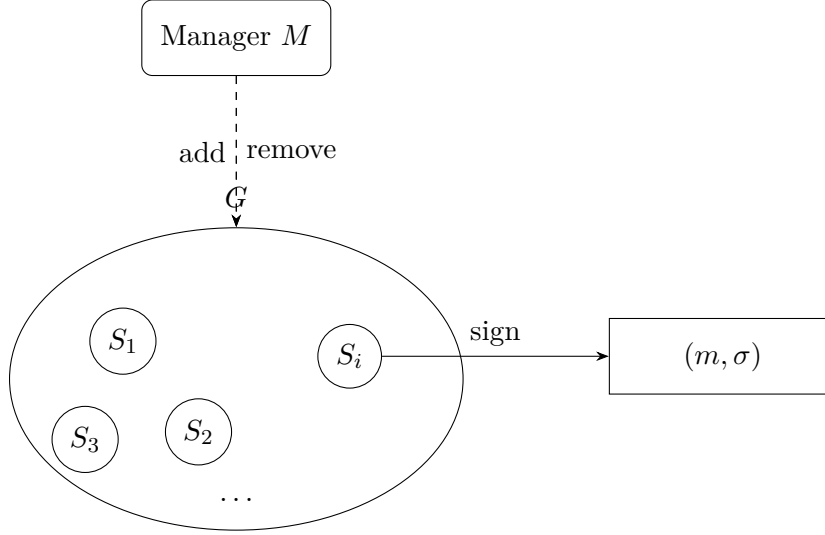
2 New PoK protocol - PoK RootLog

- Public parameters: g, e, y
- Goal: Prover knows x such that: $y = g^{(x^e)}$, where $\text{ord}(g) = N$ is an RSA modulus.
- Two parties: $\text{Prover}(x)$ and $\text{Verifier}(g, N, e, y)$ who know all public parameters.
 1. The Prover chooses $k \in_R 1, \dots, 2^\lambda$, $t = g^{(k^e)}$, where λ is a security parameter. The commitment t is sent to the Verifier.
 2. The Verifier chooses $r \in_R \{0, 1\}$ and sends it to the Prover.
 3. The Prover checks: $s = \begin{cases} k, & \text{if } r = 0 \\ \frac{k}{x}, & \text{if } r = 1 \end{cases}$, where x is the to-be-proved value of the Prover, and sends s to the Verifier.
 4. The Verifier checks: $t \stackrel{?}{=} \begin{cases} g^{(s^e)}, & \text{if } r = 0 \\ y^{(s^e)}, & \text{if } r = 1 \end{cases}$
- Correctness: $r = 0 \checkmark$
 $r = 1 : y^{(s^e)} = y^{(\frac{k}{x})^e} = g^{(x^e) \cdot (\frac{k}{x})^e} = g^{(k^e)} = t$
- Success probability: $\frac{1}{2}$, so to achieve negligible probability that the Prover succeeds with proving an incorrect value, we need to iterate the protocol several times.

2.1 Non-interactive version

- For $i = 1, \dots$: $t_i = g^{(k_i)^e}$ with random k_i , $r = H(m||y||g||e||t_1||t_2||\dots)$
- $s_i = \begin{cases} k_i, & \text{if } r_i = 0 \\ \frac{k_i}{x}, & \text{if } r_i = 1 \end{cases}$, where $r = r_1||r_2||r_3||\dots$
- $\text{PoK } x : y = g^{(x^e)}(m) = (r, s_1, s_2, \dots)$

3 Group signatures continued



- Setup:
 - M computes:
 - * $(N, c), (N, d)$ RSA keys
 - * A prime p and $g \in \mathbb{Z}_p^*$ such that $\text{ord}_p(g) = N$
 - * $a \in \mathbb{Z}_N^*$ with large order
 - * Public key: (p, g, N, e, a) , secret key: d
- Join:
 - x_i random
 - $y_i = a^{x_i} \pmod{N}$
 - $z_i = g^{y_i}$, where z_i is called the membership key
 - (y_i, z_i) is sent to M with $\text{PoK}[\alpha : a^\alpha = y_i]$
 - M returns $v_i = (y_i + 1)^d$
 - M stores y_i in $\mathcal{Y}$
- Sign:
 - S_i computes:
 - * $r \in_R \mathbb{Z}_N$
 - * $\tilde{g} = g^r$ (blinded g)
 - * $\tilde{z} = \tilde{g}^{y_i}$
 - * $V_1 = \text{PoK}[\alpha : \tilde{z} = \tilde{g}^{a^\alpha}](m)$
 - * $V_2 = \text{PoK}[\beta : \tilde{z} \cdot \tilde{g} = \tilde{g}^{\beta e}](m)$
 - * Signature = $\tilde{g}, \tilde{z}, V_1, V_2$
- Open:
 - Given $(\tilde{g}, \tilde{z}, V_1, V_2)$

- Let $\mathcal{Y}$ be the list of y_i values
- Then, check: $\tilde{g}^{y_i} \stackrel{?}{=} \tilde{z}$ for all $y_i \in \mathcal{Y}$
- Correctness:
 - By V_1 , $\tilde{g} \cdot \tilde{z}$ has the form $\tilde{g}^{y_i+1}$
 - By V_2 , S_i knows $(y_i + 1)^d = V_i$

4 Elliptic Curve Cryptography (ECC)

4.1 Motivation

- Diffie-Hellman key exchange
- Two users, A, B , they have secret keys a, b , and public keys $g^a, g^b \pmod{p}$
- It is secure if the discrete log problem is hard
- Having index calculus p must have bit size 3000-4000 (almost one megabyte)
- Using elliptic curves, we can have sufficiently secure key sizes of 200-500 bits

Definition 4.1 (Elliptic curve). An elliptic curve E over $\mathbb{R}$ is a set of points:

$$E = E_{A,B} = \{(x, y) : y^2 = x^3 + Ax + B\} \cup \{\infty\}$$

with $A, B \in \mathbb{R}$, such that $4A^3 + 27B^2 \neq 0$.

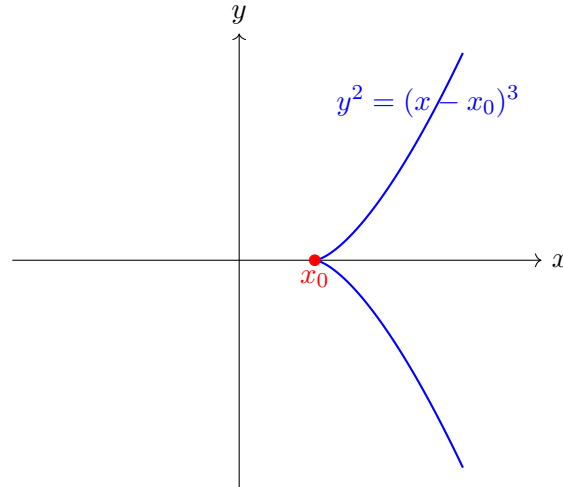


Figure 1: Singular cubic with a triple root translated to $x = x_0$.

Case 2: Normal elliptic curve

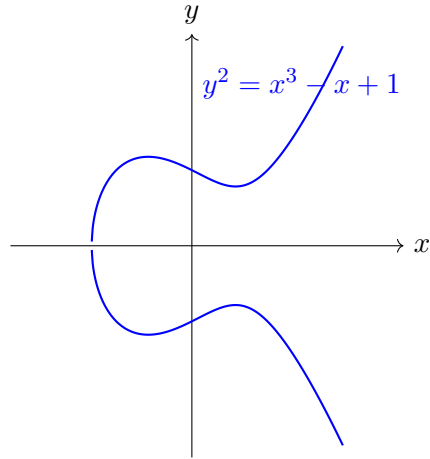


Figure 2: A normal (nonsingular) elliptic curve over $\mathbb{R}$.

Remark 4.1. E is symmetric to the x -axis: $(x, y) \in E$ if and only if $(x, -y) \in E$.

$f(x) = x^3 + Ax + B$ is cubic

Case 1: 3 different zeros

Case 2: 1 zero

- $D(A, B) = 4A^3 + 27B^2$ is the discriminant of the cubic $f(x) = x^3 + Ax + B$
- $f(x)$ has multiple zeros if $D(A, B) = 0$
- Case 1: $D(A, B) = 0, f(x) = (x - x_0)^3$
- Case 2: $D(A, B) = 0, f(x) = (x - x_0)^2(x - x_1)$

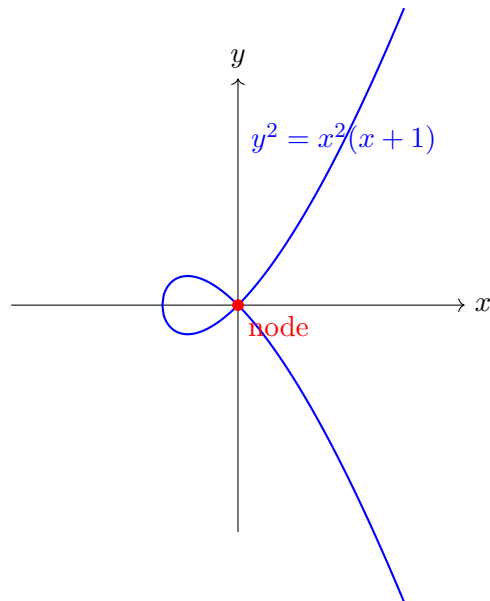


Figure 3: Singular cubic with a double root: a node.

4.2 Addition law of elliptic curves

- Let $P, Q \in E$, we define $P \oplus Q \in E$ (where $\oplus$ is the coordinate-wise addition) by the tangent-and-core method
- If $P = Q \rightarrow P \oplus Q = 2P$
- Point at infinity: basically the intersection of all vertical lines in the space. It is used when two points have the same y coordinate. The line between these two points reaches the point at infinity. In this case: $P \oplus Q = \infty, P \oplus \infty = P$

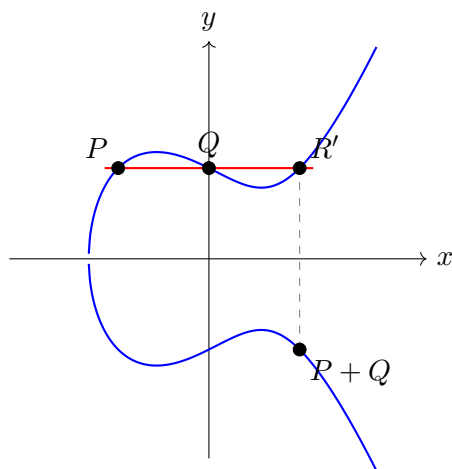


Figure 4: Geometric addition on the elliptic curve $y^2 = x^3 - x + 1$: the line through P and Q meets the curve again at R' , and $P + Q$ is the reflection of R' across the x -axis.

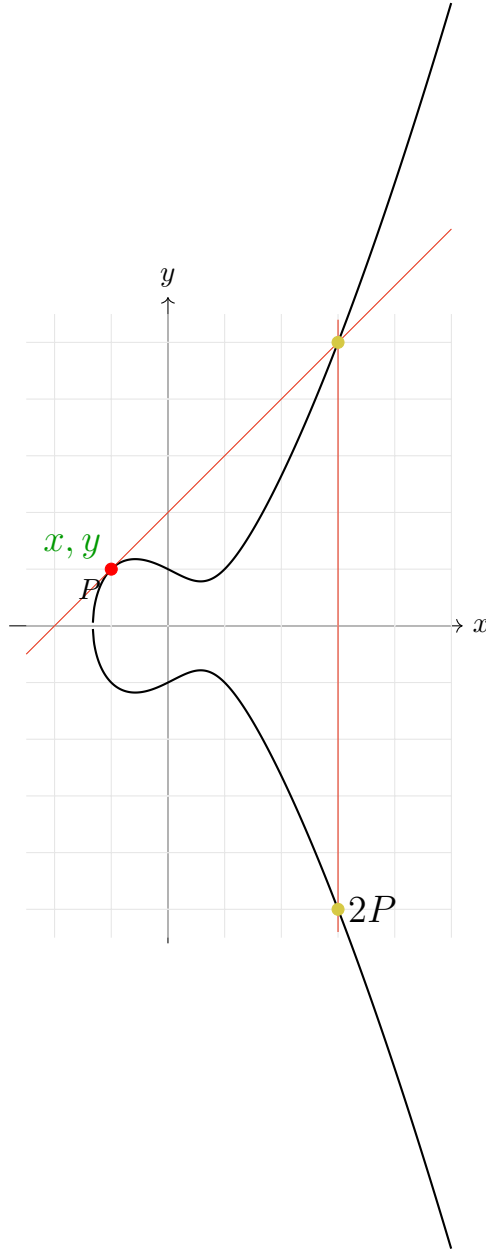


Figure 5: Point doubling on the elliptic curve $y^2 = x^3 - x + 1$: the tangent at $P = (-1, 1)$ meets the curve again at $R = (3, 5)$, and $2P = (3, -5)$ is the reflection of R across the x -axis.